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A Hybrid Synthetic Oversampling and Federated Learning Framework for Class-Imbalanced Intrusion Detection in Heterogeneous Networks

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***Abstract*—Intrusion Detection Systems (IDS) are essential for protecting modern network environments from the growing number of sophisticated cyber threats. However, the effectiveness of conventional intrusion detection models is often limited by two major challenges: severe class imbalance and heterogeneous data distributions across decentralized networks. These issues make it difficult to accurately identify minority attack classes and reduce the overall reliability of the models. To address these challenges, this paper presents a hybrid federated intrusion detection framework that combines three synthetic oversampling techniques, namely SMOTE, SMOTE-Tomek, and GAN-based SMOTE, with six federated learning algorithms: FedAvg, FedProx, FedDyn, SCAFFOLD, DP-Fed, and FedOpt. The proposed framework is evaluated using both the CIC-UNSW-NB15 and CICIDS2017 datasets individually, as well as a merged dataset, to examine the impact of dataset diversity on intrusion detection performance. Experimental results show that integrating synthetic oversampling with federated learning improves minority attack detection, enhances model robustness, and achieves better generalization while preserving data privacy. Overall, the proposed approach provides a scalable and privacy-aware solution for intrusion detection in distributed network environments. Future work will focus on validating the framework through deployment in real-world network infrastructures.**

***Index Terms*—Intrusion Detection, Class Imbalance, SMOTE-Tomek, Federated Learning, Cybersecurity.**

I. Introduction

In recent years, the rapid expansion of digital communication and interconnected network infrastructures has significantly increased the exposure to sophisticated cyber threats. As a result, Intrusion Detection Systems (IDS) have become a fundamental component in modern cybersecurity frameworks, enabling the identification of malicious activities and abnormal network behaviors. However, the performance of traditional IDS models is often constrained by the characteristics of real-world datasets, particularly the issue of class imbalance. Benchmark datasets such as UNSW-NB15 and CICIDS2017 contain a disproportionate distribution of normal and attack instances, where minority attack classes are significantly underrepresented. This imbalance leads to biased learning, causing models to achieve high overall accuracy while failing to effectively detect rare but critical intrusion types [1].

To address this issue, various data-level and algorithm-level techniques have been proposed in recent studies. Among them, SMOTE has emerged as a widely adopted approach for generating synthetic samples of minority classes, thereby improving classifier sensitivity toward underrepresented attacks. Moreover, hybrid resampling methods such as SMOTE-Tomek and SMOTE-ENN have been introduced to further enhance data quality by combining oversampling with noise reduction. Although these approaches have demonstrated improved performance in centralized environments, their applicability remains limited in distributed and privacy-sensitive scenarios, where sharing raw network data is often restricted [2].

In this context, Federated Learning has recently gained significant attention as a privacy-preserving distributed learning framework. Federated learning enables multiple clients or organizations to collaboratively train a global model without exchanging raw data, making it highly suitable for intrusion detection across decentralized network environments. However, integrating federated learning with imbalanced intrusion datasets introduces additional challenges, as data distributions across clients are often heterogeneous and skewed. This heterogeneity can further degrade the detection capability of minority attack classes, limiting the overall effectiveness of the system [3].

The key contributions of this study have been summarized as follows:

- This study investigates how synthetic oversampling and federated learning can solve the intrusion detection class imbalance problem in dispersed network systems.
- A comparative examination of CIC-UNSW-NB15, CICIDS2017, and their merged version is conducted to analyze how dataset diversity affects federated intrusion detection model performance.
- A single experimental framework is used to evaluate the performance of six federated learning algorithms:

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FedAvg, FedProx, FedDyn, SCAFFOLD, DP-Fed, and FedOpt.
- To enhance the framework's interpretability, SHAP and LIME are employed to clarify model predictions and uncover key intrusion detection aspects.

Federated intrusion detection for real-world networks with class imbalance and data diversity is the goal of this research. This study used CICIDS2017 and CIC-UNSW-NB15 benchmarks. For quality and consistency, each dataset was cleaned, normalized, and feature prepped separately. SMOTE, SMOTE-Tomek, and GAN-based SMOTE improved minority attack class representation by post-preprocessing imbalance handling. Combining the two datasets made the training and evaluation dataset more realistic and diversified. For distributed model training without client data sharing, the study employs FedAvg, FedProx, FedDyn, SCAFFOLD, DP-Fed, and FedOpt federated learning algorithms Determining minority attack class detection framework precision, recall, and F1-score. In the final section, dataset diversity, synthetic oversampling, and federated learning increase intrusion detection system reliability and generalization in individual and combined datasets.

The rest of this paper is organized as follows. Section II reviews the related literature and highlights the research gap that motivates this study. Section III describes the proposed methodology, covering data preprocessing, synthetic oversampling techniques, and the federated learning framework. Section IV presents the experimental setup and discusses the obtained results. Finally, Section V concludes the paper and provides recommendations for future work.

II. LITERATURE REVIEW

This section gives a short overview of the work that focuses on intrusion detection systems, class imbalance handling methods, synthetic oversampling tech- niques, and federated learning-based approaches

Alhussein et al. [4] reported that conventional machine learning classifiers, such as Random Forest and Support Vector Machine, achieve good performance on balanced datasets but struggle under class-imbalanced network traffic. Although these models provide high overall accuracy, they often misclassify minority attack classes, reducing their effectiveness in real-world intrusion detection systems where rare attacks are critical.

To address the class imbalance problem, Chawla et al. [5] introduced the Synthetic Minority Oversampling Technique (SMOTE), which generates synthetic minority samples instead of duplicating existing ones. This approach improves class representation and enhances the learning capability of machine learning models on imbalanced datasets. Building upon this idea, Batista et al. [6] proposed SMOTE-Tomek, which combines synthetic oversampling with Tomek Link removal to reduce class overlap and eliminate noisy samples. Their hybrid approach improves class separability and enhances the generalization performance of intrusion detection models.

McMahan et al. [7] introduced Federated Learning (FL), which enables multiple clients to collaboratively train a shared global model without exchanging raw data. This framework preserves data privacy while reducing communication costs, making it suitable for privacy-sensitive cybersecurity applications. Yang et al. [8] further showed that federated learning faces significant challenges under non-IID data distributions, where statistical heterogeneity across clients can slow convergence and reduce model accuracy. Their findings highlight the importance of designing federated learning frameworks that effectively handle heterogeneous network environments.

Sharafaldin et al. [9] introduced the CICIDS2017 dataset, which contains realistic benign and malicious network traffic and has become a widely used benchmark for intrusion detection research. Similarly, Moustafa and Slay [10] developed the UNSW-NB15 dataset to overcome the limitations of older intrusion detection datasets by incorporating modern attack categories and comprehensive network traffic features.

Paray and Aydos [11] proposed a federated intrusion detection framework that performs local oversampling before model aggregation to improve the representation of minority attack classes. Their approach enhanced both minority attack detection and global model stability. Kim and Youm [12] introduced a class-aware federated learning strategy that adapts sampling according to the local class distribution of each client, thereby improving the detection of minority attacks while maintaining the performance of majority classes.

Huang et al. [13] combined federated learning with hybrid deep learning models for IoT intrusion detection. Their framework preserved data privacy while effectively learning complex network traffic patterns, resulting in improved detection accuracy and robustness in distributed environments. Sun et al. [14] proposed FedMADE, a federated learning framework designed to address both class imbalance and non-IID client data through adaptive aggregation strategies. Their method improved the detection of minority attacks and enhanced the robustness of the global model.

Abdelkhalek and Mashaly [15] demonstrated that applying SMOTE before deep learning significantly improves the recall and F1-score of minority attack classes in imbalanced intrusion detection datasets. Talukder et al. [16] developed a machine learning-based intrusion detection system that integrates feature engineering with oversampling techniques to improve scalability and reduce false positive rates. Nawaz et al. [17] combined SMOTE with Long Short-Term Memory (LSTM) networks to enhance multi-class intrusion detection by capturing temporal dependencies in network traffic while improving the detection of minority attack classes. Sebastian et al. [18] proposed a federated intrusion detection framework for emerging environments such as the Internet of Vehicles. Their study demonstrated that imbalance-aware federated learning can simultaneously improve intrusion detection performance, privacy preservation, and scalability in distributed cybersecurity systems.

A. Research Gap

Although previous studies have demonstrated the effectiveness of synthetic oversampling techniques and federated

learning for intrusion detection, several important challenges remain. Most existing studies focus on a single oversampling method or evaluate a limited number of federated learning algorithms. In addition, comparative analyses using multiple intrusion detection datasets under a unified experimental framework are relatively limited. The interpretability of federated intrusion detection models has also received less attention, making it difficult to understand the factors influencing model predictions. To address these gaps, this study evaluates three synthetic oversampling techniques together with six federated learning algorithms using both individual and merged intrusion detection datasets. Furthermore, SHAP and LIME are incorporated to improve the interpretability of the proposed framework.

III. Methodology

This section describes the methodology adopted in this study to develop and evaluate the proposed federated intrusion detection framework. It begins with the problem formulation, followed by dataset preparation and preprocessing. The methodology then explains the synthetic oversampling techniques used to address class imbalance and the federated learning framework employed for distributed model training. Finally, the performance evaluation process is presented to assess the effectiveness of the proposed approach.

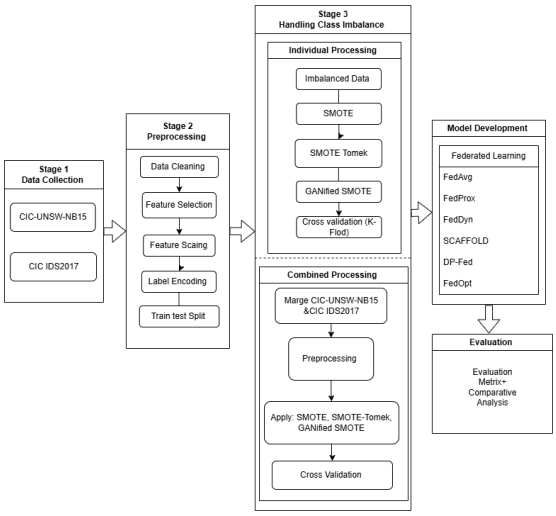


Fig. 1: Overall Workflow of Proposed Model

A. Problem Formulation

Intrusion detection systems often face two major challenges in real-world network environments: class imbalance and data heterogeneity. In most intrusion detection datasets, normal traffic significantly outnumbers malicious traffic, causing learning models to become biased toward majority classes and reducing their ability to detect minority attacks. In addition, variations in network traffic across different environments make it difficult for a single model to generalize effectively.

Although federated learning enables collaborative model training without sharing raw data, non-IID data distributions and local class imbalance remain significant challenges in distributed environments.

$$w^{t+1} = \sum_{k=1}^{K} \frac{n_k \cdot \beta_k}{\sum_{j=1}^{K} n_j \cdot \beta_j} w_k^{t+1} \tag{1}$$

where $\beta_k = \frac{\text{Minority Class Count}_k}{\text{Total Local Samples}_k}$. This ensures that client nodes experiencing severe minority class skews receive a higher relative aggregation weight. To address these issues, this study combines synthetic oversampling techniques with federated learning. SMOTE, SMOTE-Tomek, and GAN-based SMOTE are used to improve minority attack representation, while FedAvg, FedProx, FedDyn, SCAFFOLD, DP-Fed, and FedOpt are evaluated to improve distributed intrusion detection performance across heterogeneous network datasets.

B. Data Description

This study uses the CIC-UNSW-NB15 and CICIDS2017 datasets, two widely used benchmark datasets for intrusion detection research. Both datasets contain normal and malicious network traffic and were processed independently to evaluate the proposed federated learning framework. Since the minority attack classes are underrepresented, the datasets exhibit a significant class imbalance.

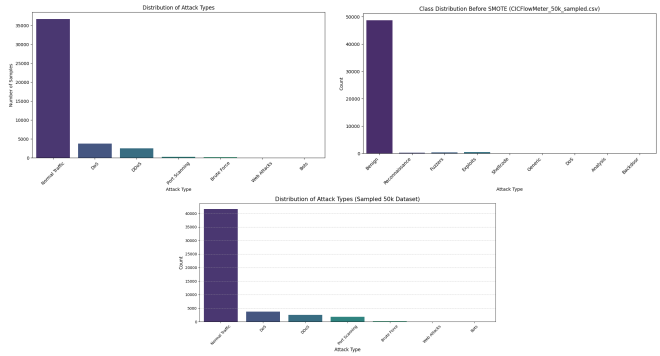


Fig. 2: Before balancing the Data

C. Preprocessing

Data was thoroughly preprocessed before model training to increase quality and consistency. The CIC-UNSW-NB15 and CICIDS2017 datasets were preprocessed individually, and the merged dataset underwent the same stages. To decrease noise, missing values and duplicate records were deleted, and label encoding turned categorical attributes into numerical values. Data normalization and learning efficiency were improved by feature scaling. These preprocessing methods provided data consistency, helping models learn from both the individual and merged datasets.

1) Phase I: Data Cleaning and Noise Removal: : The first phase aims to improve the quality of the raw network traffic data by eliminating inconsistencies and unnecessary noise. As the datasets were collected from different network environments, they include missing values, duplicate records, and

irregular entries that can affect model performance. To address these issues, missing values were handled appropriately, and duplicate records were removed to create a cleaner and more consistent dataset for model training.

- Missing Value Handling: Missing and inconsistent values were identified and handled to improve data quality and ensure reliable model training.
- Duplicate Removal: Duplicate records were removed to eliminate redundant traffic instances and reduce bias during model training.

2) Phase II: Feature Preparation and Encoding: After cleaning, the next phase prepares the dataset features for effective model training.

- Feature Selection: Irrelevant and less important features were removed to reduce dimensionality and improve model efficiency.
- Label Encoding: Categorical features, especially the target variable "Attack Type," were converted into numerical format using label encoding.

3) Phase III:Feature Scaling and Normalization: In this phase, feature scaling was applied to ensure that all numerical values remain within a similar range, improving model stability and learning efficiency.

- Normalization: The normalization process scales the feature values between 0 and 1, which can be expressed as:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{2}$$

4) Phase IV:Dataset Structuring and Splitting: The final phase prepares the dataset for model training and evaluation.

- Train-Test Split: The dataset was divided into training and testing sets using stratified sampling to maintain the original class distribution.
- Individual and Merged Dataset Processing: All preprocessing steps were first applied to the individual datasets (CIC-UNSW-NB15 and CIC IDS2017). Afterward, the datasets were merged, and the same preprocessing pipeline was applied again to ensure consistency. This allows a fair comparison between individual dataset performance and merged dataset performance.

5) Phase V:Class Imbalance Handling: After preprocessing, class imbalance was addressed using different sampling techniques.

- SMOTE (Synthetic Minority Oversampling Technique): SMOTE was applied to generate synthetic samples for minority attack classes by interpolating between neighboring data points. This improves the model's ability to learn rare attack patterns.
- SMOTE-Tomek: After generating synthetic samples, Tomek Links were used to remove overlapping and noisy samples between classes. This helps improve class separation and reduces misclassification.
- GAN-based SMOTE: Generative models were used to produce more realistic synthetic samples that capture complex data distributions, improving the robustness of the model.

D. Model Architecture

1) Federated Learning Framework: A federated learning approach lets numerous dispersed clients train a global model without exchanging raw network traffic data in the proposed intrusion detection system. This method enhances data privacy and enables secure distributed learning across networks.

2) Federated Optimization Algorithms: To comprehensively evaluate the proposed intrusion detection framework under heterogeneous federated environments, six representative federated learning algorithms were employed. Each algorithm adopts a distinct optimization strategy to address client heterogeneity, communication efficiency, convergence stability, or privacy preservation.

- **FedAvg:** FedAvg performs weighted averaging of locally trained client models to update the global model.

$$w^{t+1} = \sum_{k=1}^{K} \frac{n_k}{n} w_k^t \tag{3}$$

  where w_k^t denotes the local model parameters of client k, n_k represents the number of training samples at client k, and $n = \sum_{k=1}^{K} n_k$. FedAvg serves as the baseline federated optimization algorithm because of its simplicity and communication efficiency.
- **FedProx:** FedProx extends FedAvg by introducing a proximal regularization term that constrains excessive deviation between the local and global models.

$$\min_w F_k(w) + \frac{\mu}{2}\|w - w_t\|^2 \tag{4}$$

  where μ controls the strength of the proximal regularization. This constraint reduces client drift and improves convergence under heterogeneous data distributions.
- **FedDyn:** FedDyn introduces dynamic regularization to align local optimization objectives with the global objective.

$$F_k(w) - \langle y_k, w\rangle + \frac{\alpha}{2}\|w\|^2 \tag{5}$$

  where y_k is the accumulated dual variable and α controls the regularization strength. FedDyn improves optimization consistency and accelerates convergence across heterogeneous clients.
- **SCAFFOLD:** SCAFFOLD mitigates client drift by employing control variates during local optimization.

$$w_k^{t+1} = w_k^t - \eta(\nabla F_k - c_k + c) \tag{6}$$

  where c_k and c denote the local and global control variates, respectively. This mechanism significantly improves optimization stability under non-IID client distributions.
- **DP-Fed:** DP-Fed incorporates Differential Privacy into federated learning by perturbing gradients before aggregation.

$$g' = g + \mathcal{N}(0, \sigma^2) \tag{7}$$

where Gaussian noise is added to clipped gradients before global aggregation. Although this slightly reduces predictive performance, it effectively protects sensitive client information during communication.

- **FedOpt:** FedOpt employs adaptive server-side optimization during global aggregation.

$$w^{t+1} = w^t - \eta_t m_t \tag{8}$$

where m_t denotes the adaptive optimization direction computed at the server. This adaptive optimization strategy improves convergence speed and enhances training stability across heterogeneous clients.

3) Training on Individual and Merged Datasets: The federated learning framework was evaluated using both individual datasets and the merged dataset.

- Individual Dataset Training: Helps analyze dataset-specific learning behavior under distributed environments.
- Merged Dataset Training: Provides more diverse attack patterns and improves model generalization across heterogeneous network traffic.

4) Performance Evaluation: The effectiveness of the proposed federated intrusion detection framework was evaluated using standard classification metrics, particularly the F1-score, due to the highly imbalanced nature of intrusion detection datasets.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{9}$$

IV. Experiment and Result

A. Experimental Settings

The experiments were conducted in a cloud-based environment with GPU support. Python libraries, including Pandas, NumPy, TensorFlow, and Scikit-learn, were used for data preprocessing, synthetic oversampling, federated learning implementation, and performance evaluation. Visualization tools were used to examine dataset distributions and analyze model performance.

B. Imbalance Handling on Individual and Merged Datasets

Different synthetic oversampling methods, including SMOTE, SMOTE-Tomek, and GAN-based SMOTE, were applied, shown in TableI and Figure3 across all datasets to improve the representation of minority attack classes.

TABLE I: F1-score Comparison of Different Imbalance Handling Techniques on Individual and Merged Datasets

Dataset	SMOTE	SMOTE-Tomek	GAN-based SMOTE
Dataset 15	0.9988	0.9989	0.9988
Dataset 17	0.9996	0.9997	0.9996
Merged Dataset	0.9925	0.9911	0.9989

The results indicate that all three oversampling techniques improve the representation of minority attack classes while reducing the bias toward majority classes. SMOTE enhances minority class distribution through synthetic sample generation, SMOTE-Tomek further improves class separation by removing overlapping samples, and GAN-based SMOTE generates more realistic synthetic attack samples for effective model learning.

TABLE II: Parameter Settings Used in Federated Learning Experiments

Parameter	Value
Federated Algorithms	FedAvg, FedProx, FedDyn, SCAFFOLD, DP-Fed, FedOpt
Communication Rounds	10
Batch Size	32
Evaluation Metrics	Precision, Recall, F1-score
Train-Test Split	80:20

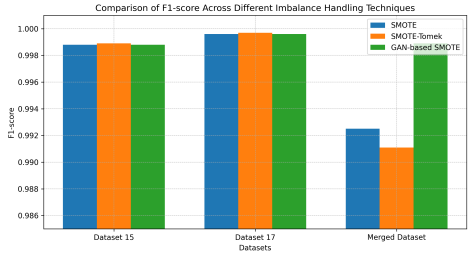


Fig. 3: Comparison of F1-score Across Different Imbalance Handling Techniques

C. Performance of Federated Learning

The proposed federated intrusion detection framework was evaluated using the individual datasets (CIC-UNSW-NB15 and CICIDS2017) as well as the merged dataset to examine the impact of data diversity and imbalance handling on intrusion detection performance. FedAvg, FedProx, FedDyn, SCAFFOLD, DP-Fed, and FedOpt were used for distributed model training without sharing raw network data. The results demonstrate improved minority attack detection and enhanced model generalization across heterogeneous network environments.

The framework was trained using the parameter settings presented in Table II to ensure a fair comparison across the individual and merged dataset scenarios.

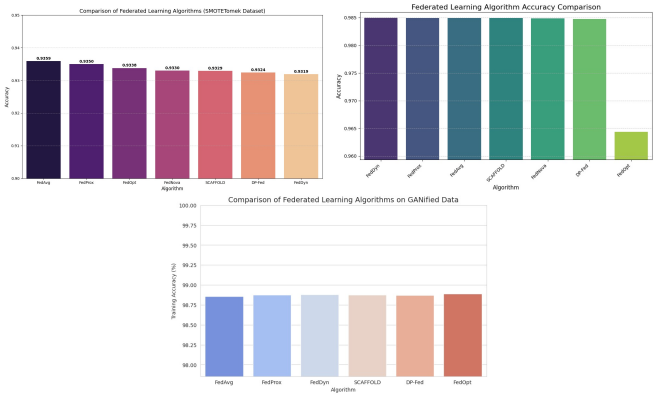


Fig. 4: Perfomance of Federated Learning

The results presented in Fig4 show that the performance of the federated learning framework varies across differ-

ent datasets and balancing techniques. While the individual datasets capture dataset-specific traffic characteristics, the merged dataset provides greater diversity in attack patterns and network behavior. This contributes to improved model generalization and more effective detection of minority attack classes in distributed learning environments.

D. Interpritability of Performance

SHAP and LIME were used to interpret the predictions of the proposed federated learning model on the merged CIC-UNSW-NB15 and CICIDS2017 dataset. SHAP provides a global explanation of feature importance by identifying the features that contribute most to intrusion detection. LIME complements this analysis by explaining individual predictions, particularly for minority attack classes. Together, these methods improve the transparency of the proposed framework and support a better understanding of the model's decision-making process.

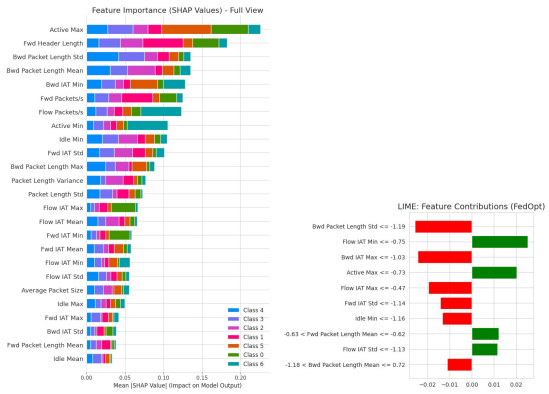


Fig. 5: Shap and Lime(FedOpt) on Merged Dataset

E. Discussion

The experimental results demonstrate that both dataset diversity and class imbalance handling influence the performance of federated intrusion detection systems. Compared with the individual datasets, the merged dataset provides a wider range of attack patterns and network traffic characteristics, which improves the model's ability to generalize across heterogeneous environments. Among the evaluated federated learning algorithms, FedOpt achieved the highest accuracy on the merged dataset, while FedAvg and FedDyn also showed competitive performance on the individual datasets. In addition, the application of SMOTE, SMOTE-Tomek, and GAN-based SMOTE improved the representation of minority attack classes. These findings indicate that combining synthetic oversampling with federated learning can enhance intrusion detection performance in heterogeneous network environments.

V. Conclusion

This study proposed a federated intrusion detection framework to address class imbalance and data heterogeneity using the CIC-UNSW-NB15 and CICIDS2017 datasets. SMOTE, SMOTE-Tomek, and GAN-based SMOTE were applied to improve the representation of minority attack classes, while six federated learning algorithms were evaluated under a common experimental framework. The experimental results demonstrate that combining synthetic oversampling with federated learning improves minority attack detection and enhances model robustness across heterogeneous network environments. Furthermore, SHAP and LIME improve the interpretability of the proposed framework by explaining both global feature importance and individual predictions. Future work will focus on validating the proposed framework in real-world distributed environments and exploring additional privacy-preserving optimization techniques.

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